

Designing Graceful Degradation for Multimodal Autonomous Vehicle Cabins

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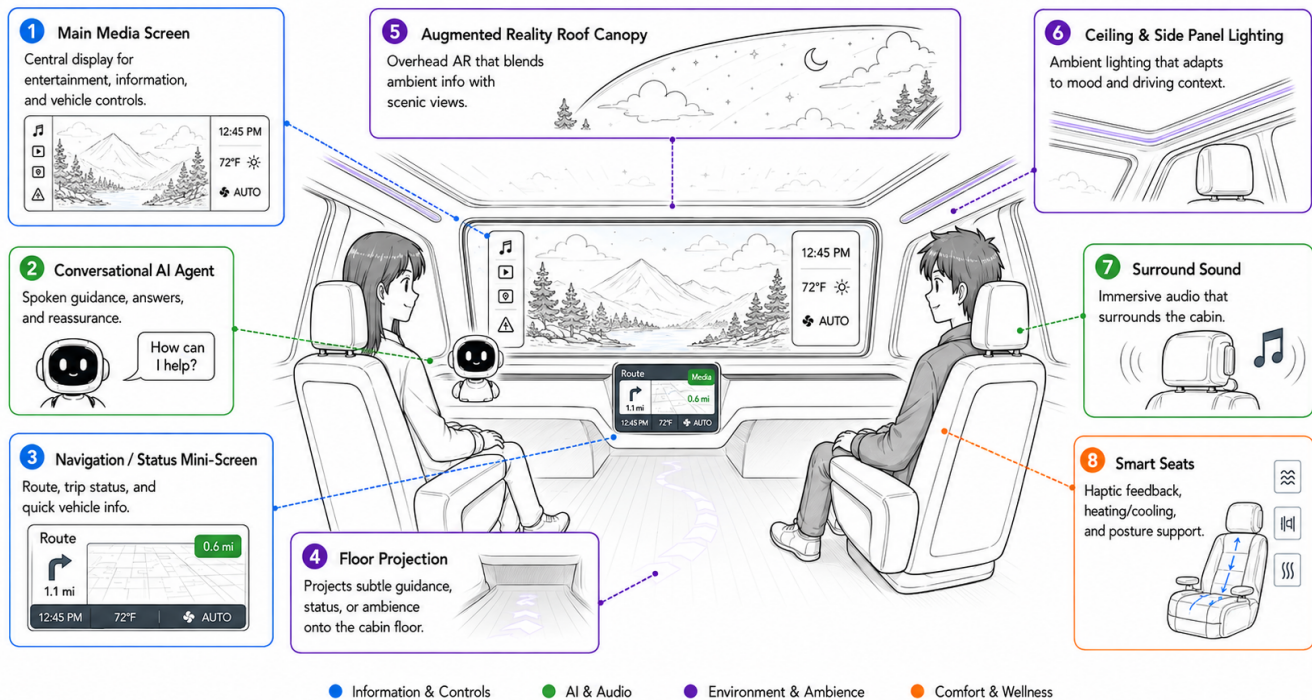


Figure 1: Concept illustration of the autonomous vehicle cabin and the multimodal interface elements considered in our design space.

Abstract

Future autonomous vehicle cabins may function as immersive spaces in which riders rest, work, or watch media while the vehicle manages navigation and control. Hazards create a communication challenge because riders may have little awareness about road conditions when the cabin needs to convey risk. We examine *graceful degradation* as a design lens for coordinating multimodal cabin responses across changing hazard states. We conducted a storyboard-based study with $N = 12$ participants who reviewed three scenarios

with different time profiles: a gradual snowstorm, an intermediate pedestrian event, and a sudden incoming vehicle. Participants reflected on staged visual, auditory, and haptic cues. They associated graceful responses with cue strength that tracked hazard tempo, visible system intent, restrained haptics, and a prompt return to the baseline cabin state after danger passed. We position graceful degradation as a temporal interaction framework for rider-facing Level 5 autonomous vehicle cabins. We also identify design tensions and next-step studies around escalation thresholds, haptic intensity, sensory load, trust, and emergency cue offset.



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AutomotiveUI Adjunct '26, Gothenburg, Sweden
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ACM ISBN 979-8-4007-2815-0/26/09
<https://doi.org/10.1145/3828158.3838226>

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); Scenario-based design.

Keywords

autonomous vehicles, human-machine interfaces, multimodal cues, graceful degradation, scenario-based design

ACM Reference Format:

Yumeng Ma and Rani Malhotra. 2026. Designing Graceful Degradation for Multimodal Autonomous Vehicle Cabins. In *18th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI Adjunct '26)*, September 20–23, 2026, Gothenburg, Sweden. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3828158.3838226>

1 Introduction

Autonomous vehicles (AVs) are commonly described through a future in which riders disengage from driving and use travel time for rest, entertainment, or work. In Level 5 automation, the vehicle performs the driving task without human intervention [14, 16]. A rider may therefore encounter heavy snow, a pedestrian, an incoming vehicle, or another hazardous condition with little attention directed toward the road. The cabin must communicate changing risk to someone who may be watching media, sleeping, or otherwise detached from the driving scene.

Current automotive interfaces use visual, auditory, and haptic cues to communicate hazards or changes in automation state [5, 7, 8, 20, 22–24]. Much prior research examines drivers who retain some responsibility for monitoring or control. Rider-only cabins pose a different interaction problem because riders cannot intervene through steering or braking. We use *graceful degradation* to examine how cabin cues can communicate changing system state across escalation and recovery. Graceful degradation originates in fault-tolerant computing [6, 15], and recent AV research applies the concept to decision and control responsibilities [1]. We extend this framing toward rider-facing interaction by examining how multimodal cues appear, intensify, coordinate, and recede as risk changes.

We conducted a storyboard-based study with 12 participants who reviewed three speculative hazard scenarios with different time profiles: a gradual snowstorm, an intermediate pedestrian event, and a sudden incoming vehicle. Participants reflected on visual, auditory, and haptic cue sequences across each scenario. To our knowledge, this is the first study to apply and operationalize *graceful degradation* for multimodal, rider-facing Level 5 AV cabin human-machine interface (HMI) design. Our findings characterize how riders reason about cue strength, vehicle intent, haptics, sequencing, and recovery across hazard tempos. We contribute this rider-facing framing, empirical findings across three hazard tempos, and design directions for future automotive research.

2 Background

2.1 Graceful Degradation and AV Fallback

Graceful degradation has a long history in computing, distributed systems, and fault-tolerant engineering. The central idea is that a system continues in a reduced, usable state when full operation fails [6, 15]. Previous interface research applies related principles to multiplatform layouts and transition continuity [4, 9, 21]. AV research has started to apply graceful degradation to control and decision responsibilities. Aniculaesei et al. propose a verifiable method that redistributes decision and control responsibilities when

automated driving functions degrade [1]. This systems-oriented framing concerns how the vehicle preserves safe operation. Our work addresses how riders experience that transition when they do not hold driving responsibility. A vehicle may manage an event appropriately yet create an abrupt or frightening rider experience when the cabin does not communicate what changed or what will happen next. We treat graceful degradation as both a vehicle-state transition and a cabin communication problem.

2.2 Multimodal Warnings and Automation Transparency

Multimodal warning research shows that visual, auditory, and haptic cues can support attention and response when they align with hazard location, timing, and urgency [5, 7, 8, 24, 26]. Most studies assess a driver's response to a warning, takeover request, or mode change. The central outcome can concern reaction speed, takeover quality, gaze, or workload. Additionally, automation transparency research examines how people interpret system confidence, perception, and intent. Confidence indicators, explanations, and trajectory visualizations can support trust calibration and situational awareness [2, 10–12, 25]. These techniques can also add clutter or cognitive demand during time-sensitive events [3, 13, 19]. Graceful degradation contributes a temporal flow across these areas. It organizes when existing modalities appear, how they combine, how strongly they escalate, and how they withdraw after a hazard. This lens moves analysis from isolated cues toward the cabin's movement across normal, degraded, emergency, and recovered states.

3 Storyboard Concepts

We designed three storyboarded hazard scenarios around a shared speculative AV cabin. Storyboards support early discussion when implementation would be costly, risky, or premature [18]. We used them as design probes. The baseline cabin used a front media display, a navigation mini-screen, an augmented-reality roof canopy, ceiling and side lighting, floor projection, spatial audio, smart-seat haptics, seatbelt tension, and a conversational agent.

Figure 2 summarizes the three temporal profiles. The snowstorm represented gradual degradation. Light snow appeared outside and in the roof canopy. As visibility and traction worsened, the cabin lighting shifted cooler, the media display added light snow effects, and the navigation screen displayed a low-traction cruise mode. The agent explained speed and following-distance adjustments. If conditions became unsafe, entertainment paused and the vehicle entered a stopped state with route and safety information.

The pedestrian scenario represented an intermediate hazard. The vehicle detected a pedestrian near the roadside, then shifted attention toward the forward scene. The media screen introduced a semi-transparent camera overlay, the mini-screen added a pedestrian marker, and directional cues appeared through lighting, floor projection, and spatial audio. As the pedestrian entered the path, the display shifted toward a fuller camera view and presented a short safety message.

The incoming-vehicle scenario represented an immediate hazard. Another vehicle swerved toward the AV lane. The cabin introduced rapid cues across red edge highlights, directional floor pulses, spatial audio, seatbelt tension, and a forward view with the planned evasive

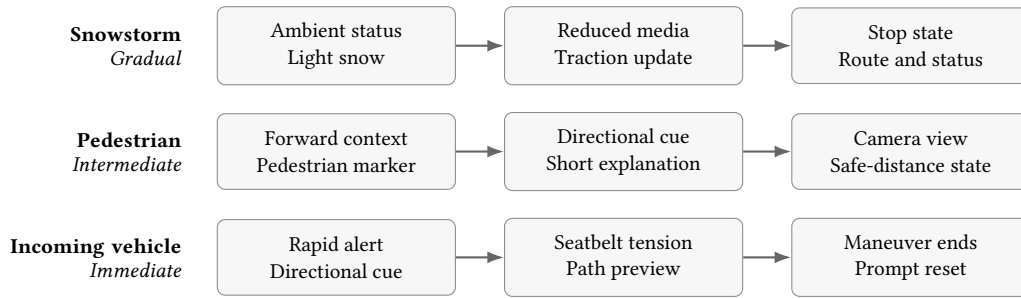


Figure 2: Temporal flow across three storyboard scenarios. Each sequence progressed through increasingly salient cabin states, with escalation speed and modality strength impacted by hazard tempo.

path. After the maneuver, the cabin returned to its baseline media state. This final transition let participants evaluate emergency offset as well as onset.

4 Study Method

4.1 Participants

We recruited 12 participants through personal networks and referrals within industry and consumer AV communities. Participants ranged in age from 22 to 54. Six identified as women and six as men. Two worked professionally in the AV sector. Three owned consumer AVs or used automated features in daily travel. One had never ridden in an AV. Two were non-drivers. The remaining participants drove conventional vehicles and had no AV ownership experience. All participants were frequent commuters or car-service users. We note that recruitment may have favored people willing to discuss speculative AV concepts and may not completely represent riders who distrust automation, experience motion sensitivity, or have access needs that change how they perceive multimodal cues.

4.2 Procedure

Each remote session lasted about 45 minutes through Microsoft Teams or Zoom. Participants first saw the baseline cabin and then reviewed all three scenarios in counterbalanced order. The facilitator narrated each sequence to preserve consistent pacing and explain visual, auditory, and haptic changes that static frames could not convey. Participants could add optional notes to a shared FigJam board. After each scenario, participants described what felt graceful, disruptive, reassuring, confusing, or excessive. They discussed cue timing, modality combinations, system intent, and return-to-baseline behavior. After all scenarios, participants described how they understood graceful degradation in an AV cabin and proposed other emergency contexts for future study.

4.3 Analysis

The first author recorded detailed session notes, supplemented by FigJam annotations. We used an inductive coding process [17]. The first author developed an initial code set covering hazard tempo, escalation sequence, system intent, haptic acceptance, visual takeover, outside visibility, and recovery. The second author reviewed the coded material and category boundaries. Both authors refined the

final themes through consensus. Table 1 presents representative codes and examples.

Table 1: Representative codes used during analysis.

Code	Interpretation
Tempo match	Cue intensity fits how quickly risk develops
Visible intent	Interface shows what the vehicle detects and plans
Intrusive haptic	Touch-like cue feels startling, invasive, or unnecessary
Escalation order	Modalities appear through a legible sequence
Recovery signal	Cabin marks hazard resolution and returns to baseline
Outside grounding	Rider retains access to road or environmental context

5 Findings

Our storyboards encouraged participants to judge graceful degradation through the relationship among hazard tempo, cue strength, system intent, and recovery. Participants evaluated how the cabin transitioned across states.

5.1 Cue Strength Tracked Hazard Tempo

Participants expected stronger cues as time-to-risk shortened. The snowstorm prompted preferences for calm pacing, low-traction status, speed-adjustment explanations, and periodic updates. P8 described the sequence as “*gradual and did not create too much alarm.*” P2 argued that once the vehicle communicated deceleration, extra effects were unnecessary: “*Floor projection changes feel a bit too much.*”

The incoming-vehicle scenario generated greater acceptance for strong cues. Participants supported seatbelt tension, centering pressure, spatial sound, and a forward camera view because the maneuver required rapid interpretation and bodily preparation. P7 said, “*Seatbelt tension and seat centering here are very effective.*” P9 preferred maximal escalation during the short emergency window: “*Use everything that you have to bring me to that alert state.*”

The pedestrian scenario produced mixed reactions. Participants liked early forward context and directional cues but disagreed about full camera takeovers, fading transitions, and haptics. P3 suggested a short banner before a media takeover. P5 preferred a fast visual transition because a slow fade increased nervousness. Intermediate hazards may therefore demand tighter timing thresholds than slow or immediate events.

5.2 Riders Wanted the Vehicle's Plan

Participants wanted the cabin to communicate both detection and planned action. Hazard labels named a problem, but path traces, pull-over arcs, and brief speech helped participants understand what would happen next. During the snowstorm, participants requested information about traction, stopping, route changes, and progress during a long hold. P9 wanted updates about *"range, steps toward a safe pull-out,"* and other status metrics. During the incoming-vehicle event, participants valued the lane trace because it explained the evasive maneuver. P3 said the sequence from peripheral dimming to the forward view felt clear and that the lane trace supported spatial awareness. This pattern suggests that rider-facing transparency may be most useful when it conveys action.

5.3 Haptics Were Accepted Near Physical Action

Participants reacted strongly to narrated haptic concepts. Seatbelt tension and centering pressure gained support during the incoming-vehicle event because they aligned with bodily preparation. Touch-like cues gained less support. P5 responded to seat movement by asking, *"Why are my shoulders being hit? Why is my seat being moved?"* P10 explained, *"I really don't like haptics—I don't even like it when people touch me."*

Participants also rejected strong haptics during slow hazards. Smart-seat pressure during the snowstorm felt more like restraint than reassurance. These reactions do not establish how implemented haptics would feel because force, rhythm, location, posture, and anticipation could change the response. Haptics gained acceptance when they corresponded to immediate vehicle motion and lost acceptance when they served as general notification.

5.4 Sequencing Impacted Multimodal Coherence

Several participants preferred cues to build in stages in comparison to appearing simultaneously. P9 described a sequence that began with a primer and then introduced visual information. Across accounts, a recurring arc emerged: a low-intensity cue signals change, a short label names the event, a plan display shows vehicle intent, a bodily cue prepares the rider near action, and a closure signal marks recovery. This sequence does not prescribe one modality at each stage. Graceful degradation organizes modalities according to the rider's changing need for orientation, explanation, preparation, and closure.

5.5 Grace Extended Through Emergency Offset

Participants evaluated how the cabin left an emergency state. Lingering red light, persistent bounding boxes, and delayed media recovery prolonged tension after the event. P3 described the *"hazard averted"* cue as *"a nice emotional reset."* Other participants preferred

short closure language such as "clear" followed by a quick return to the normal cabin. Cue offset may influence trust, anxiety, and perceived recovery. A cabin can communicate that a hazard has ended through a closure cue, the withdrawal of emergency visuals, and a stable transition back to baseline. A delayed or ambiguous reset can imply that danger continues after vehicle behavior stabilizes.

6 Discussion

6.1 Graceful Degradation Adds Temporal Pacing

Warning research compares modalities, combinations, and takeover timing. Transparency research examines explanations, confidence, and predicted action. Graceful degradation adds a state-transition perspective across these concerns. It asks how a cabin moves from routine travel into reduced capability, how cue strength changes as uncertainty develops, and how the interface closes the event.

For immersive Level 5 cabins, riders cannot rely on steering control to interpret vehicle intent. Participants evaluated whether the cabin established a legible plan, escalated in step with time-to-risk, and withdrew emergency cues after stability returned. Graceful degradation is a pacing logic for coordinating displays, lighting, audio, and haptics across a hazard trajectory.

The findings also clarify limits. Not every hazard permits gradual staging. Tire blowouts, high-speed intrusions, or stacked failures may require an abrupt response. Grace does not mean calmness at every moment. It concerns a response whose pace and force correspond to the hazard and whose state changes can be interpreted by riders.

6.2 Design Directions

Table 2 translates the findings into preliminary design directions and testable next steps. The recommendations are provisional because participants reacted to storyboards instead of implemented systems.

Table 2: Preliminary design directions and next-step investigations.

Direction	Next-step investigation
Match cue strength to hazard tempo	Test escalation thresholds across gradual, intermediate, and immediate hazards
Show system intent early	Compare detection labels with path previews and short action explanations
Reserve haptics for physical preparation	Test intensity, location, rhythm, timing, and opt-out controls
Preserve outside grounding	Compare direct sightlines, camera views, and augmented overlays
Design onset and offset together	Test how emergency entry and reset timing affect trust, anxiety, and recovery

The next phase could center on AV-naïve riders and non-drivers, followed by groups whose sensory, cognitive, or mobility access needs may alter cue perception.

6.3 Prompts for Further Investigation

When should a cabin enter and leave emergency alerting? Simulator studies can test time-to-collision, uncertainty, speed, road conditions, and rider activity as inputs to escalation. They can also vary how quickly warning cues withdraw after stabilization and measure trust, anxiety, perceived recovery, and resumption of prior activity.

How should systems communicate uncertainty without creating panic? A vehicle may detect deteriorating conditions before it knows whether emergency action will be needed. Interfaces need forms that distinguish monitoring, constrained operation, and immediate intervention.

Which haptics support readiness without violating rider comfort? Physical prototypes should test force, body location, rhythm, anticipation, and rider control. Haptics may require firm safety policies plus adjustable comfort settings.

When should immersion recede? Participants sometimes preferred straightforward road context over layered effects. Research can compare windows, camera views, simplified safety displays, and augmented overlays during each hazard phase.

7 Limitations

Our study used remote storyboards and not an implemented cabin, simulator, or physical prototype. Participants imagined sound, lighting, haptics, and motion through narration and static visuals. Findings about sensory load and bodily response should be treated as reactions to design concepts. We do note that the small sample, recruited through personal networks and referrals cannot universally claim about Level 5 riders or subgroup comparisons. Further, the cabin concept used more modalities than a production system would likely activate together. This breadth helped expose preferences and rejection points. However, future prototypes should test smaller modality sets under controlled timing and intensity.

8 Conclusion

We examined graceful degradation as a temporal design lens for multimodal Level 5 AV cabin interfaces. Twelve participants reviewed three storyboarded hazards with gradual, intermediate, and immediate time profiles. They judged graceful responses through cue strength that tracked hazard tempo, visible vehicle intent, haptics tied to physical action, legible sequencing, and a prompt reset after danger passed. These findings distinguish rider-facing graceful degradation from control-level fallback and from isolated warning design. The next research step is higher-fidelity evaluation with AV-naïve riders and non-drivers, followed by studies across rider access needs and sensory preferences.

Acknowledgments

We thank Crystal George for making this opportunity possible, our participants for their valuable contributions, and the Infosys Center for Emerging Technology Solutions for supporting this research.

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